

The Cyclic Nucleotide Phosphodiesterase Inhibitory Protein of *Dictyostelium discoideum*

PURIFICATION AND CHARACTERIZATION*

(Received for publication, August 25, 1980, and in revised form, March 16, 1981)

Jakob Franke and Richard H. Kessin

From the Department of Cellular and Developmental Biology, The Biological Laboratories, Harvard University, Cambridge, Massachusetts 02138

We have purified the glycoprotein inhibitor of the extracellular cyclic nucleotide phosphodiesterase of *Dictyostelium discoideum* to apparent homogeneity. The inhibitor has a molecular weight of 47,000 measured by sodium dodecyl sulfate-polyacrylamide gel electrophoresis. The interaction of the inhibitor and the cyclic nucleotide phosphodiesterase occurs with 1:1 stoichiometry and with a dissociation constant of about 10^{-10} M. Periodate oxidation of the inhibitor or of the enzyme destroys concanavalin A binding ability but does not affect the formation of the enzyme-inhibitor complex. Inhibitor is not produced by cells during logarithmic growth but appears in quantity during stationary phase and after transfer from growth medium to phosphate buffer.

When cells of the social amoeba *Dictyostelium discoideum* are removed from nutrient medium, they develop the ability to detect, secrete, and move toward sources of cAMP. During the first 8–10 h of starvation, the cells produce a number of proteins thought to be necessary for chemotaxis and cellular adhesion (see Ref. 1 for a review). Among the proteins synthesized are an extracellular cyclic nucleotide phosphodiesterase and a glycoprotein which specifically inhibits the phosphodiesterase.

The extracellular cyclic nucleotide phosphodiesterase and its inhibitor are postulated to control the levels of cAMP during aggregation, the enzyme by degrading cyclic nucleotide between pulsatile signals and the inhibitor by regulating levels of phosphodiesterase activity so that the cAMP signal is not attenuated. Amoebae regulate the level of enzyme and inhibitor according to the extracellular cAMP concentration; high levels of cAMP stimulate enzyme appearance and suppress that of the inhibitor (2–5). Inhibitor appears in the extracellular supernatant of cAMP-treated cells once the cAMP has been degraded by the action of the phosphodiesterase.¹ A further reason for believing that the cyclic nucleotide phosphodiesterase and the inhibitor play important roles during aggregation is that all species of slime mold which utilize cAMP as a chemotactic signal elaborate an extracellular cyclic nucleotide phosphodiesterase and inhibitor, while those which use other molecules for signaling do not (6).

* This work was supported by Grant GM21,809 and a Research Career Development award (to R. H. K.) from the National Institutes of Health. The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

¹ Unpublished results.

Neither the regulation of the inhibitor by cAMP nor the role it plays during chemotaxis and later stages of development can be understood until the molecule is purified. When the current study began, the inhibitor, which was discovered by Riedel and Gerisch (7), was known to be heat-stable and trypsin-sensitive (8). The molecule had been partially purified and shown to be a glycoprotein (9). More recently the inhibitors from *D. discoideum* and *D. purpureum* were purified (10, 11), although in neither case was enough material recovered for complete biochemical characterization.

In initial experiments with partially purified inhibitor we determined that the kinetic effect of the inhibitor is to alter the K_m of the phosphodiesterase from 10 μ M to 2 mM. The association of enzyme and inhibitor is disrupted by dithiothreitol, which inactivates the inhibitor (12, 13). The current paper presents methods for the production and purification of inhibitor as well as more evidence on the nature of the inhibitor and its interaction with cyclic nucleotide phosphodiesterase. The purification of cyclic nucleotide phosphodiesterase is described in the accompanying paper (14).

MATERIALS AND METHODS²

RESULTS

Cellular Synthesis of Cyclic Nucleotide Phosphodiesterase Inhibitor

Growing cells have secreted between 100 and 200 units/ml of inhibitor by the time they reach stationary phase in HL/5 medium (Fig. 1A). Fig. 1B shows the result of transferring the cells to starvation medium shortly after they reached stationary phase. If the cells were left in stationary phase for another 48 h, they continued to secrete inhibitor until a level of 400–600 units/ml was reached. Little extracellular phosphodiesterase was secreted. Cells left at stationary phase for 48 h and then washed and placed in buffer to starve produced over 600 units/ml of inhibitor, but never any significant levels of enzyme (Fig. 1C). Therefore, the same set of cells produced 600 units/ml of inhibitor while in late log and stationary phase and another 600 units/ml when transferred to buffer.

For preparative purposes cells were harvested at the beginning of stationary phase and starved for 13 h in phosphate

² Portions of this paper (including "Materials and Methods," some of the "Results," Figs. 1 and 2, Tables 1–3, and additional references) are presented in miniprint at the end of this paper. Miniprint is easily read with the aid of a standard magnifying glass. Full size photocopies are available from the Journal of Biological Chemistry, 9650 Rockville Pike, Bethesda, MD 20014. Request Document No. 80M-1827, cite author(s), and include a check or money order for \$4.00 per set of photocopies. Full size photocopies are also included in the microfilm edition of the Journal that is available from Waverly Press.

buffer plus CaCl_2 (see "Materials and Methods"). Cells treated in this way produced about 500 units/ml of inhibitor. Inhibitor recovered from the supernatants of these cells had a higher specific activity than inhibitor recovered from stationary phase supernatants. The values described above were for the preparative cultures described under "Materials and Methods." When the cell densities of cells starving in buffer were lowered 2- to 4-fold, the inhibitor levels were proportionally reduced.

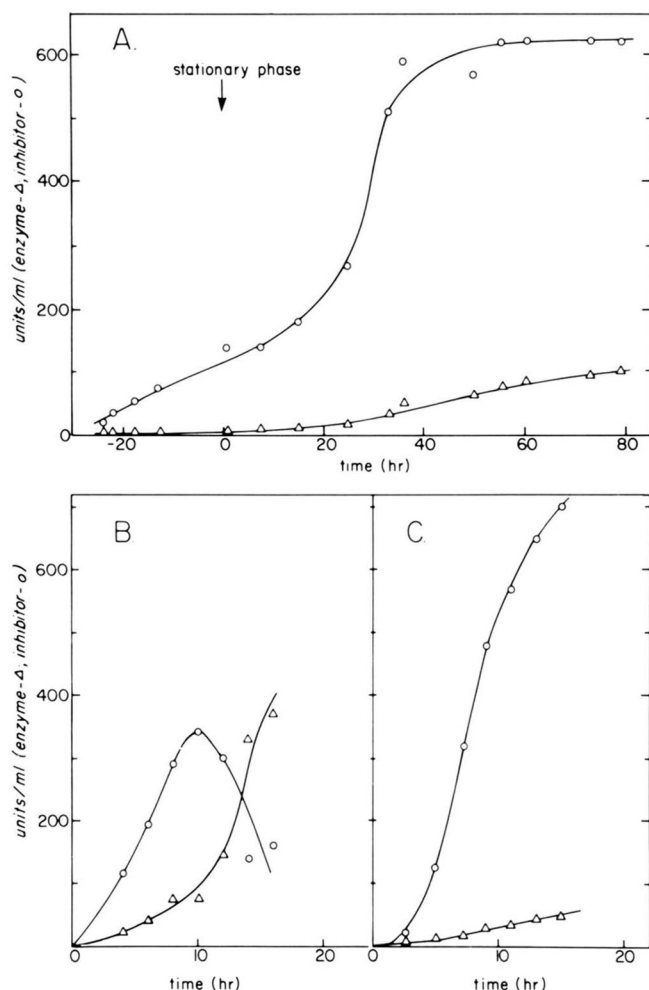


FIG. 1. The production of cyclic nucleotide phosphodiesterase and inhibitor during growth, stationary phase, and starvation. A, 700-ml cultures of AX3 in HL/5 were incubated as described under "Materials and Methods." At various times aliquots were removed and centrifuged (5 min; $200 \times g$). The pH of the supernatants was measured, and the samples were stored at -20°C . The cell density in stationary phase was 2.6×10^7 cells/ml. The pH increased from 7.3 (onset stationary phase) to 8.4 (80-h stationary phase). Aliquots were dialyzed for 48 h against two changes of Tris (50 mM Tris-HCl; pH 7.4 at 35°C) or 24 h against 2 mM dithiothreitol in Tris, followed by 24 h against Tris containing 0.1 mM dithiothreitol. The dialyzed samples were assayed for phosphodiesterase activity by the spectrophotometric assay method. Tris-dialyzed samples contained no detectable phosphodiesterase activity (<1 unit/ml) and were assayed for inhibitor activity following heating for 15 min at 80°C . Δ , enzyme activity; $\circ$, inhibitor activity. The ordinate shows a time scale in which the 0 point indicates the intercept of the exponential growth phase curve and the stationary phase curve. B, cultures of AX3 in HL/5 were harvested 4 h after the onset of stationary phase and starved in phosphate buffer as described under "Materials and Methods." Samples were harvested and assayed as described in A. C, cultures of AX3 in HL/5 were harvested 61 h after the onset of stationary phase and starved in buffer as in B.



FIG. 2. SDS-polyacrylamide gel electrophoresis of purified inhibitor. 3000 units of inhibitor were subjected to electrophoresis in 11% SDS gels and stained with Coomassie brilliant blue as described under "Materials and Methods."

Criteria of Homogeneity

The absence of contaminants was demonstrated by SDS³-polyacrylamide gel electrophoresis. The inhibitor was stained with Coomassie brilliant blue (Figs. 2 and 4) as well as with glycoprotein stains and in neither case were contaminants detected. The glycoprotein migrated as a diffuse band or as a

³ The abbreviations used are: SDS, sodium dodecyl sulfate; Tris or Tris buffer, 50 mM Tris-HCl (pH 7.4 at 35°C); BSA, bovine serum albumin; BIS, *N,N'*-methylenebisacrylamide; TEMED, *N,N,N',N'*-tetramethylethylenediamine; EGTA, ethylene glycol bis(β -aminoethylether)*N,N,N',N'*-tetraacetic acid.

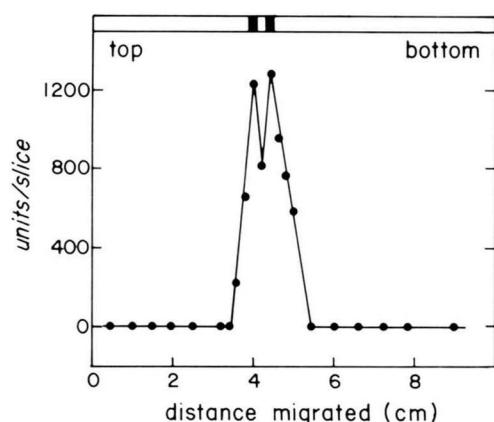


FIG. 3. Electrophoresis of inhibitor under nondenaturing conditions. A 5% polyacrylamide slab gel with spacer gel was run using the pH 8.9 buffer system (see "Materials and Methods"). Gels were fixed and stained for 4 h at room temperature in 0.2% Coomassie brilliant blue R250 in H₂O, methanol, acetic acid (5:5:1), and destained at 50 °C in 7% acetic acid containing 0.5% methanol. Lanes loaded with 8500 units of inhibitor were cut into 2-mm slices. Each slice was eluted overnight at 4 °C in 1 ml Tris, and the eluate was assayed for inhibitor activity; 70–80% of the inhibitor was recovered.

doublet in SDS gels (see also Fig. 4 of the accompanying paper (14)). When purified inhibitor was subjected to electrophoresis under nondenaturing conditions, only two bands were detected, although these sometimes merged into a single band. Glycoprotein stains (Alcian blue and periodic acid-Schiff) did not reveal any additional bands. When native gels were sliced and eluted, two peaks of inhibitor activity were found at positions corresponding to the stained bands (Fig. 3). Two peaks of activity were also detected when partially purified material was analyzed by column isoelectrofocusing. These peaks had isoelectric points of 3.4 and 4.3, respectively (results not shown).

Another demonstration of the purity of the inhibitor preparation was obtained by reacting dithiothreitol treated purified inhibitor with [³H]iodoacetic acid. The labeled product was analyzed by SDS-gel electrophoresis and one peak of radioactivity, corresponding to the stained bands, was observed (Fig. 4).

During electrophoresis under a variety of conditions it became apparent that the inhibitor glycoprotein was sometimes difficult to stain. In native gels Amido black stained the inhibitor as efficiently as did Coomassie brilliant blue. The same amount of protein stained more intensely in SDS gels than it did in nondenaturing gels. This effect was amplified by increasing the 2-mercaptoethanol concentration in the samples prior to SDS-gel electrophoresis.

Molecular Weight

Varying molecular weights were observed for the inhibitor during electrophoresis in 7.5% and 10% SDS-polyacrylamide gels. Fig. 5 shows the results when the apparent molecular weights were determined with a range of polyacrylamide gel concentrations (15). These results revealed a molecular weight at the asymptote of 47,000. Gel filtration also yielded varying molecular weights. On Sephacryl S-200 columns the inhibitor consistently eluted with an apparent molecular weight of about 50,000, while on an Ultrogel AcA34 column the molecular weight was 29,000 (Fig. 6) or 33,000 (see Fig. 8 of accompanying paper (14)).

Amino Acid Analysis

The results of an amino acid analysis performed on a purified preparation of inhibitor are shown in Table I. The

most noteworthy features of the amino acid composition are the high cystine content and the low level of methionine. The minimum cystine content is 5.9/100 residues, and extrapolating to 400 residues, the minimum number of half-cystine residues is 47/molecule. This is a minimum estimate because some cysteine residues are lost during hydrolysis (16).

The carbohydrate content of the inhibitor was determined with the anthrone reaction (17) using D-glucose as the standard. Inhibitor contained 0.07–0.08 mg of carbohydrate/mg of protein. Since it is known that several sugars, including mannose and fucose, have a lower color yield than glucose in the anthrone assay and hexosamines give no color at all, our estimate is probably a minimal one. A positive identification of glucosamine was made during the amino acid analysis of the 24-h hydrolysate.

Inhibitor-Cyclic Nucleotide Phosphodiesterase Interaction Stoichiometry

The strategy used to determine the stoichiometry of the inhibitor-enzyme interaction was to mix the two components with one in excess. In this way, the molecular weights of the resulting complex and one of the two reactants could be compared in one gel filtration analysis. The results are shown in Figs. 6 and 7. Inhibitor was assayed before and after heating aliquots of the column fractions; enzyme activity was determined before and after activation with dithiothreitol. Fig. 6 shows the results when 2150 units of enzyme and 2500 units

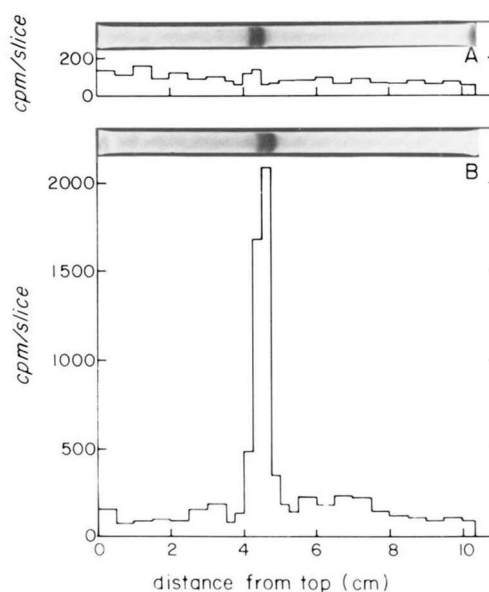


FIG. 4. SDS-polyacrylamide gel electrophoresis of ³H-labeled inhibitor. 2400 units of inhibitor (5.4 µg) in 20 mM Tris-HCl (pH 7.4 at 35 °C) with or without 2 mM dithiothreitol were heated for 2 min in boiling water. Total volume was 60 µl. After cooling, 30 µl of [³H]iodoacetate were added (20 mM [³H]iodoacetic acid in 250 mM Tris-HCl; specific activity, 19.3 Ci/mol). The samples were incubated 30 min at 35 °C in the dark, at which time 25 µl of 20% (v/v) mercaptoethanol containing 5% (w/v) SDS were added. The samples were heated 5 min in boiling water, and then sucrose and bromophenol blue were added. The samples were subjected to electrophoresis in 10% SDS-polyacrylamide gels in tubes (see "Materials and Methods"). Following electrophoresis the gels were fixed and stained with Coomassie brilliant blue as described. The destained gels were sectioned in 2.5- or 5-mm slices, which were incubated for 1.5 h at 60 °C in 2-dram vials containing 1 ml of 30% H₂O₂. The hydrolyzed slices were counted after the addition of 3.5 ml of Scintiverse. The counting efficiency was 8.1% and was determined by the addition of [³H]iodoacetic acid to vials containing unlabeled polyacrylamide slices prior to hydrolysis. A, without dithiothreitol. B, with 2 mM dithiothreitol.

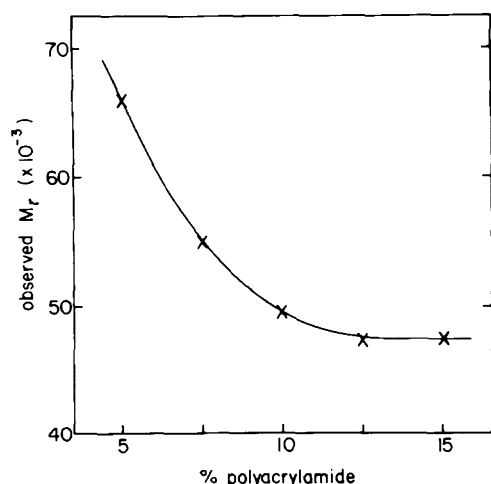


FIG. 5. Molecular weight determination of inhibitor glycoprotein by SDS-polyacrylamide gel electrophoresis. SDS gels with varying concentrations of polyacrylamide were prepared as described under "Materials and Methods." Samples were boiled with 1% SDS and 2% mercaptoethanol. All gels were run with a constant current of 2mA/tube until bromophenol blue reached the bottom of the tubes. Molecular weight markers were phosphorylase A (94,000), BSA (67,000), pyruvate kinase (57,000), ovalbumin (45,000), deoxyribonuclease I (31,000), soybean trypsin inhibitor (21,500), and ribonuclease A (13,700).

TABLE I

Amino acid composition of inhibitor

Residue	Amount mol/100 mol
Asp	13.7
Thr ^a	13.1
Ser ^a	10.0
Glu	5.3
Pro	7.0
Gly	8.0
Ala	4.3
Cys ^b	5.9
Val	3.5
Met	0.3
Ile	4.4
Leu	4.4
Tyr ^a	6.2
Phe	2.9
Lys	6.5
His	2.1
Arg	2.1

^a The values for Thr, Ser, and Tyr are extrapolations to 0-time hydrolysis.

^b The values for cystine were 4.9, 5.9, and 0.9 after 24-, 48-, and 72-h hydrolysis, respectively.

of inhibitor were fractionated on an Ultrogel AcA34 gel filtration column. Enzyme activity was nearly undetectable in fractions which had not been treated with dithiothreitol (Fig. 6A). After dithiothreitol treatment of the eluted fractions, 90% of the original enzyme activity could be recovered in a peak corresponding to $M_r = 77,000$. When the unheated eluate was assayed for inhibitor activity a peak of $M_r = 29,000$ was found (Fig. 6B), corresponding to excess free inhibitor. If aliquots of the column fractions were heated to 80 °C for 15 min to destroy phosphodiesterase and to release bound inhibitor, an additional inhibitor peak, fractionating in the same position as the enzyme activity peak appeared. A similar experiment was performed in the accompanying paper (14) where we reported a complex with an apparent $M_r = 84,000$, and excess free inhibitor with an apparent $M_r = 33,000$.

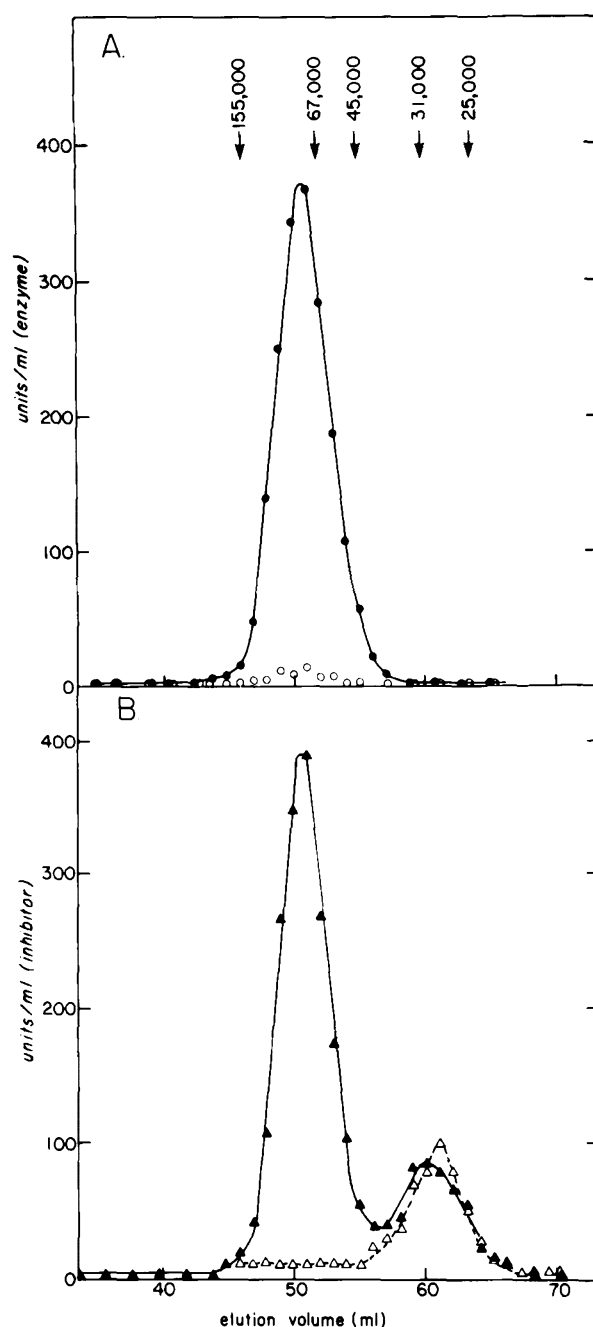


FIG. 6. Gel filtration of complex and excess inhibitor. 2150 units of purified extracellular phosphodiesterase ($M_r = 50,000$) and 2500 units of purified inhibitor were incubated for 30 min at 35 °C in a total volume of 1 ml and fractionated on an AcA34 gel filtration column (bed volume, 74.5 ml; bed height, 74 cm). The elution volume of blue dextran was 29.5 ml and was used as void volume. The flow rate was 4.12 ml/h with a constant hydrostatic head of 20 cm. Fractions of 1.03 ml were collected. The elution buffer⁴ was: Tris containing 0.2 M NaCl, 0.01% bovine serum albumin and 0.01% NaN₃. A: ○, enzyme activity without dithiothreitol activation; ●, enzyme activity following dithiothreitol activation. B: △, inhibitor activity without heating; ▲, inhibitor activity following 15 min heating at 80 °C.

⁴ In the experiments described in Figs. 6 and 7, the enzyme and the inhibitor, as well as their complex, bind to the gel matrix when the elution buffer is Tris. The inclusion of 0.2 M NaCl and 0.01% bovine serum albumin in the buffer eliminates this problem and results in 90–100% recovery.

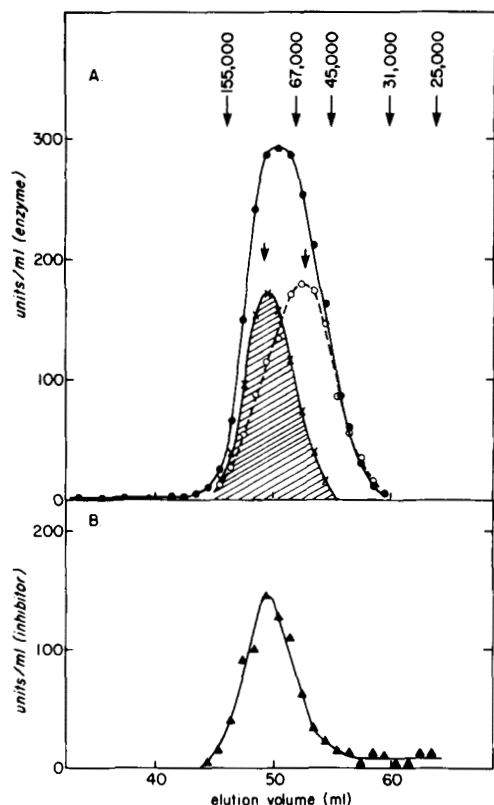


FIG. 7. Gel filtration of complex and excess phosphodiesterase. 2150 units of purified enzyme ($M_r = 50,000$) and 880 units of purified inhibitor were incubated and fractionated by gel filtration as described in the legend of Fig. 6. The flow rate was 4.0 ml/h, and 1.0-ml fractions were collected. A: $\circ$, enzyme activity without dithiothreitol activation; $\bullet$, enzyme activity following dithiothreitol activation. The hatched region corresponds to the differential curve. B: $\blacktriangle$, inhibitor activity after 15 min heating of fractions at 80 °C.

Although inhibitor was added in excess based on activity (Fig. 6), it was not in molar excess. 2500 units of inhibitor correspond to 0.12 nmol ($M_r = 47,000$ on SDS gels; specific activity, 440,000 units/mg), while 2150 units of enzyme equal 0.31 nmol ($M_r = 50,000$ by SDS-gel electrophoresis; specific activity, 140,000 units/mg; see Ref. 14). In order to account for the excess of inhibitor activity, one must postulate a complex of 3 enzyme molecules with 1 inhibitor molecule. The gel filtration data shown in Fig. 6 do not support such a large complex with a calculated molecular weight of 197,000. The observed size of the complex (77,000–84,000) indicates a stoichiometry of 1 inhibitor to 1 enzyme molecule. We believe that the high specific activity of the inhibitor is caused by an underestimate of the inhibitor protein concentration. The protein assays used display significant protein to protein variation (18), and we have noted above the difficulties encountered when staining the inhibitor glycoprotein.

The results of an experiment with enzyme in excess are shown in Fig. 7. The elution profile of cyclic nucleotide phosphodiesterase activity assayed with and without destruction of the inhibitor by dithiothreitol is shown (Fig. 7A). Free enzyme eluted at a position corresponding to a molecular weight of 59,000. Treatment of the fractions with dithiothreitol inactivated the inhibitor and resulted in an unmasking of the enzyme. If the activity of the free enzyme was subtracted from the activity of the dithiothreitol treated enzyme, a differential curve corresponding to the enzyme-inhibitor complex was produced (hatched region, Fig. 7A). This peak, with an apparent molecular weight of 87,000 contained inhibitor activity (Fig. 7B).

When cyclic nucleotide phosphodiesterase activity was totally suppressed by excess inhibitor and then treated with dithiothreitol, the recovered enzyme activity eluted from a gel filtration column at the position of the free enzyme with an apparent $M_r = 55,000$ (results not shown). From all of these results we conclude that the enzyme has an apparent M_r by gel filtration of 55,000–59,000 and that it reacts with 1 inhibitor molecule of $M_r = 29,000$ –33,000, to form a complex of $M_r = 77,000$ –87,000.

The existence of a stable inhibitor-phosphodiesterase complex was also demonstrated by isoelectrofocusing. The results obtained following isoelectrofocusing in polyacrylamide gels of phosphodiesterase, inhibitor, and complexes with either an excess of enzyme or an excess of inhibitor, are as follows: phosphodiesterase focused between pH 7.8 and pH 8.7, enzyme-inhibitor complexes focused between pH 6.1 and pH 7.8, and inhibitor focused between pH 3.8 and pH 4.9.

Dissociation Constant

The dissociation constant of the enzyme-inhibitor complex was determined by three methods. In each case we made the assumption that the phosphodiesterase activity measured at low substrate concentrations reflects the free enzyme concentration. Units of activity were converted to nanomoles of phosphodiesterase using the factor 7000 units/nmol of enzyme; this is based on a specific activity of 140,000 units/mg of protein and a molecular weight of 50,000 by SDS-gel electrophoresis for the phosphodiesterase (see the accompanying paper (14)).

The final specific activity we have measured for the inhibitor is 440,000 units/mg of protein. We have shown in Figs. 6 and 7 that the stoichiometry of the interaction is 1:1, so that the specific activity of the inhibitor should be similar to that of the phosphodiesterase, shown in the accompanying paper (14) to be 140,000 units/mg protein. We have attributed this difference to the difficulty of staining the inhibitor and believe that our estimate of protein may be low, thus accounting for the high specific activity. In calculating the dissociation constant we have used 7,000 units/nmol as the specific activity of purified inhibitor.

Method 1—Purified cyclic nucleotide phosphodiesterase was reacted with an excess of purified inhibitor, and the remaining concentrations of enzyme, inhibitor, and complex were determined. The data presented in Table II were used to calculate an average equilibrium constant of 1.7×10^{-10} M.

Method 2—Inhibitor and an excess of enzyme were incu-

TABLE II
Equilibrium constant of enzyme-inhibitor complex with excess inhibitor

Enzyme-inhibitor complexes were prepared by mixing aliquots of purified enzyme and inhibitor in Tris. The samples were incubated 1 h at 35 °C, and stored several days at 4 °C. The free enzyme activity was assayed by the radiometric method, and total enzyme was determined spectrophotometrically following treatment with 2 mM dithiothreitol (see "Materials and Methods"). Total inhibitor activity was assayed after heating aliquots 15 min at 80 °C in the presence of 0.2 M NaCl. The concentration of complex was taken as the average value of (total enzyme – free enzyme) and (total inhibitor – free inhibitor). $K = [\text{free enzyme}][\text{free inhibitor}]/[\text{complex}]$. We have used a specific activity of 7000 units/nmol for the enzyme and for the inhibitor (see the text).

Experiment	Free enzyme	Total enzyme	Free inhibitor	Total inhibitor	K
	units/ml				M
1	7.2	210	80	250	4.4×10^{-11}
2	0.62	250	260	460	1.0×10^{-10}
3	0.47	230	170	330	0.6×10^{-10}
4	1.5	550	240	780	0.9×10^{-10}

bated at 35 °C at different concentrations but in a constant ratio and the remaining enzyme activities were determined. In this case, enzyme was in excess, so any contribution to cyclic nucleotide hydrolysis by the complex would be negligible. The concentration of complex was calculated as total enzyme minus free enzyme and the concentration of free inhibitor was taken as total inhibitor minus complex. The dissociation constant was calculated from the equation:

$$K = \frac{(E)(I)}{(EI)} = \frac{E_t(I_t - E_t + E_f)}{(E_t - E_f)}$$

where E_t , I_t , and E_f represent the total enzyme, total inhibitor, and free enzyme concentrations, respectively. Using this method over a 30-fold range of concentrations and the conversion factors given above, dissociation constants of 9×10^{-11} , 1×10^{-11} , and 1.1×10^{-10} were derived. Data are not shown.

Method 3—In the ΔI method described by Greco and Hakala (19) the dissociation constant can be calculated using the equation $K = (2 \times I_{75}/3) - I_{50}$ where I_{75} and I_{50} are the total inhibitor concentrations at 75 and 50% inhibition, respectively. Increasing amounts of inhibitor were mixed with 3.5 units of enzyme under the conditions of the inhibitor assay (see "Materials and Methods"). Following a 30-min incubation at 35 °C the remaining enzyme activities were determined. The results, plotted in the same way as those in Fig. 1 (see miniprint section), were used to calculate a dissociation constant of 2×10^{-11} M (data not shown).

The rate of complex formation was very fast. Inhibitor (150 units/ml) and enzyme (90 units/ml) were mixed at 35 °C, and at 15-s intervals the remaining enzyme activities were measured over a 30-s period with the radiometric assay. The inhibition was 80% complete within 45 s, 90% complete within 60 s, and 99% complete within 15 min. When the inhibition of enzyme (22 units/ml) with inhibitor (27 units/ml) was followed in the presence of 50 μ M cAMP, the rate of complex formation was substantially reduced. After 60 s, the inhibition was only 20–30% complete.

Once formed, the complex is stable in the presence of cAMP. Complex (230 units/ml of enzyme; 330 units/ml of

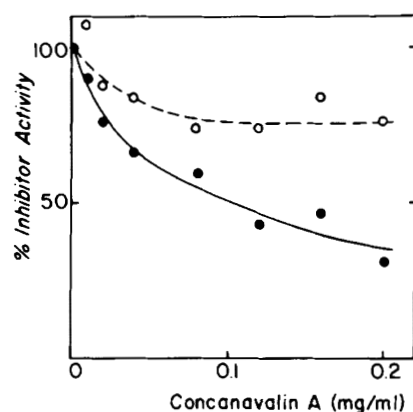


FIG. 8. The effect of concanavalin A on inhibitor-enzyme interaction in the presence and absence of α -methylmannoside. Five units of cyclic nucleotide phosphodiesterase were incubated with 3 units of purified inhibitor in Tris buffer in the presence of varying amounts of concanavalin A. The total volume was 1.32 ml and incubations were for 15 min at 35 °C. The remaining enzyme activities were assayed spectrophotometrically after the addition of 25 μ l of coupling enzymes (see "Materials and Methods") and 50 μ l of 1 mM cAMP to 925 μ l of incubation mixture. ●, decreasing inhibitor activity with increasing concentration of concanavalin A (100% inhibitor activity corresponding to about 40% remaining enzyme activity); ○, the same experiment but in the presence of 0.2 M α -methylmannoside.

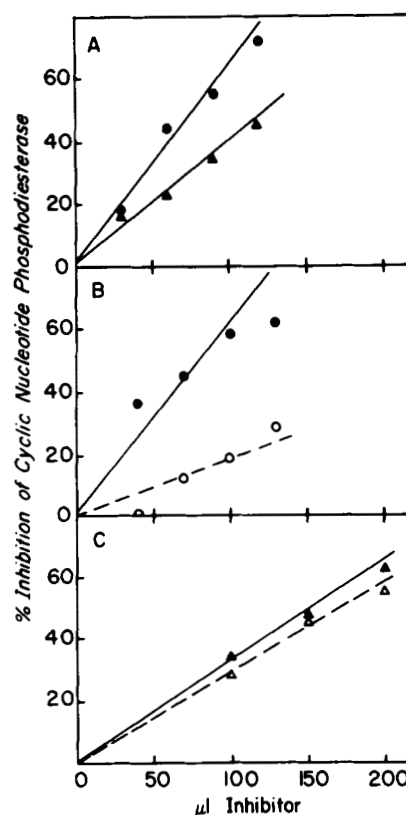


FIG. 9. The effect of periodate oxidation of cyclic nucleotide phosphodiesterase on its interaction with the inhibitor. Five units of enzyme were incubated with varying amounts of inhibitor, and the remaining enzyme activities were determined as described in the legend of Fig. 8. The periodate oxidation of the enzyme has been described in the legend of Table III. Oxidized enzyme had lost its ability to bind to concanavalin A-Sepharose 4B and had about 60% of the activity of untreated enzyme. A, the inhibition of untreated enzyme (●) and periodate-oxidized enzyme (▲); B, the inhibition of untreated enzyme in the absence (●) or presence (○) of concanavalin A (0.2 mg/ml); C, the inhibition of periodate oxidized enzyme in the absence (▲) or presence (△) of concanavalin A (0.2 mg/ml).

inhibitor) was assayed with two substrate concentrations in the radiometric assay. With 50 μ M cAMP, the activity was 0.47 units/ml and at 1 mM substrate concentration, the activity was 10 units/ml. In both cases, the hydrolysis of cAMP was linear for at least 30 min, demonstrating that the complex is not dissociating during an incubation of 30 min in the presence of 1 mM cAMP. We cannot rule out a very rapid partial dissociation of the complex after the addition of 1 mM cAMP.

Attempts were made to measure the rate of dissociation. Enzyme-inhibitor complex was prepared at high concentration and percent inhibition was determined (about 50%). Following a 100-fold dilution and incubation for several hours, percentage inhibition did not change.

Oligosaccharides and the Enzyme-Inhibitor Interaction

Since both the inhibitor and the enzyme are glycoproteins, it is possible that the formation of the enzyme-inhibitor complex involves a lectin type of binding. The effect of the following sugars on the formation of the complex was tested: D-glucose, D-galactose, D-mannose, D-fucose, L-fucose, D-glucosamine, N-acetyl-D-glucosamine, N-acetyl-D-galactosamine, α -methyl-D-mannoside, α -methyl-D-glucoside, myoinositol, and α -lactose. At a final concentration of 0.2 M, none of these sugars had any effect on either the enzyme activity or the inhibitor activity.

Concanavalin A has no effect on cyclic nucleotide phospho-

diesterase activity, but the interaction of the enzyme with the inhibitor is affected by concanavalin A, as shown in Figs. 8 and 9B. The presence of 0.2 M α -methylmannoside counteracts this effect (Fig. 8).

In order to determine whether the oligosaccharide residues of the inhibitor or of the enzyme participate in the enzyme-inhibitor interaction, we treated the purified glycoproteins with periodate. As the results in Table III show, periodate-treated inhibitor no longer binds to concanavalin A, but retains its ability to inhibit cyclic nucleotide phosphodiesterase. Periodate oxidation at pH 4.5 destroys 80% of the inhibitor activity (Table III, experiment 1). This is not surprising since the amino acids tryptophan, cysteine, cystine, and methionine can be oxidized by periodate, especially at low pH (20). Periodate treatment at pH 6.0 eliminates binding to concanavalin A with less loss of inhibitor activity (Table III, experiment 2).

When periodate-treated inhibitor was assayed in the presence of concanavalin A, the lectin still interfered with the formation of the enzyme-inhibitor complex (Table IV). This indicates that it is not the binding of lectin to the inhibitor, but its reaction with the enzyme which interferes with complex formation. To demonstrate more directly that it is concanavalin A binding to the enzyme which blocks complex formation, periodate-oxidized cyclic nucleotide phosphodiesterase was prepared. The oxidation was again monitored by

TABLE III

Periodate treatment of inhibitor

Periodate oxidations were performed in a total volume of 4.0 ml containing 50 mM Na-acetate, pH 4.5 (experiment 1) or 50 mM Na-citrate, pH 6.0 (experiment 2), 10 mM periodic acid, and inhibitor. The samples were incubated overnight at 4 °C in the dark, and the reaction was terminated by the addition of 1.0 ml of 0.2 M glucose. One-ml aliquots were applied to a concanavalin A-Sepharose 4B column (0.4-ml bed volume; 1-cm bed height) equilibrated in Tris buffer. After washing with Tris, the bound inhibitor was eluted with 20% α -methylmannoside in Tris. Bound and unbound inhibitor are expressed as percentages of the units applied.

Experiment	pH	Condition	Inhibitor	Concanavalin A-Sepharose 4B	
				Unbound	Bound
			units/ml	%	
1	4.5	- periodate	1100	0	94
		+ periodate	240	65	26
2	6.0	- periodate	1300	0	77
		+ periodate	980	90	0

TABLE IV

Effect of concanavalin A on inhibitor treated with periodate

Periodate treatments in experiments 1 and 2 were performed in acetate buffer (pH 4.5), while in experiment 3 citrate buffer (pH 6.0) was used (see the legend to Table III). Approximately 3 units of inhibitor were incubated with 5 units of cyclic nucleotide phosphodiesterase in Tris buffer with and without concanavalin A (0.2 mg/ml). The total volume was 1.32 ml. Following a 30-min incubation at 35 °C the remaining enzyme activities were determined spectrophotometrically after the addition of 50 μ l of 1 mM cAMP and 25 μ l of coupling enzymes (see "Materials and Methods") to 925 μ l of the incubation mixture. The results are expressed as the per cent inhibition of controls containing no inhibitor.

Experiment	Inhibition of phosphodiesterase			
	Inhibitor (-IO ₄ ⁻)		Inhibitor (+IO ₄ ⁻)	
	- Con A	+ Con A	- Con A	+ Con A
1	51	30	57	24
2	61	23	44	18
3	52	29	51	22
Average	55	27	51	21

TABLE V

Periodate treatment of cyclic nucleotide phosphodiesterase

Periodate oxidations were performed at pH 6.0 as described in the legend of Table III. Concanavalin A binding was done in batches. One-ml aliquots were incubated 30 min at room temperature with 1 ml of concanavalin A-Sepharose 4B suspension (0.3 ml of packed beads) in Tris. Following low speed centrifugation, the supernatant was mixed with 1 ml of fresh suspension and treated as before. The pellets were combined and washed twice with 1 ml of Tris buffer. The combined supernatants and washings contained unbound enzyme, expressed as a percentage of the original units. Bound enzyme was determined by eluting the pellets twice with 2 ml of 20% (w/v) α -methylmannoside in Tris.

Conditions	Enzyme	Concanavalin A-Sepharose 4B	
		Unbound	Bound
	units/ml	%	
- periodate	285	14	66
+ periodate	203	99	8

TABLE VI

pH stability of the inhibitor

Aliquots of inhibitor were incubated at room temperature under the conditions shown. The inhibitor concentration was 600 units/ml. At the times indicated aliquots were diluted 10-fold in 0.5 M Tris buffer and assayed for remaining activity. The table shows the percentage of the original activity.

Inhibitor	Original activity after:			
	1 h	4 h	24 h	96 h
	%			
Tris	100	100	100	100
1 M acetic acid	94	90	85	87
0.36 M trichloroacetic acid	102	92	81	59
0.05 N HCl	105	102	106	93
0.7 M Na ₂ CO ₃	100	80	74	17
0.05 N NaOH	30	0	0	—

the loss of binding to concanavalin A-Sepharose 4B (Table V). Since the enzyme is not stable at low pH, the oxidations were carried out at pH 6.0. The results shown in Fig. 9A indicate that, although the periodate-treated enzyme has lost its ability to bind concanavalin A, its activity can still be depressed by the inhibitor. However, 30–40% more inhibitor is needed to achieve a given level of inhibition if the enzyme has been periodate oxidized. Since the periodate-treated enzyme has lost about 30% of its activity, the inactive enzyme may still be competing with active enzyme for the inhibitor. Alternatively, the oligosaccharide of the enzyme may have a function in inhibitor binding.

When periodate-oxidized enzyme was incubated with the inhibitor in the presence of concanavalin A, no effect of concanavalin A on the inhibitor-enzyme complex formation was observed (Fig. 9C). This again demonstrates that the binding of concanavalin A to the cyclic nucleotide phosphodiesterase interferes with complex formation.

Ion Effects

The effect of several ions and chelators on the formation of enzyme-inhibitor complex was examined in the presence of bovine serum albumin (0.1 mg/ml) using the radiometric assay. CaCl₂, MgCl₂, and NH₄Cl at 2 mM concentrations had no effect on the inhibition of enzyme activity, nor did 10 mM EDTA, 10 mM ethylene glycol bis(β -aminoethyl ether)*N,N,N',N'*-tetraacetic acid, or a mixture of the two at 5 mM each. *O*-Phenanthroline (2 mM) had no effect except to inhibit enzyme activity by about 30%; 2 mM 5'-AMP and 20 mM potassium phosphate did not interfere with the inhibition, while the presence of 0.5 M NaCl diminished inhibition by about 10%.

Inhibitor Stability

The inhibitor is not only extremely resistant to denaturation by heat (30 min at 80 °C showed no loss of activity) but is also stable at low pH. Table VI shows the remaining activity after exposure of the inhibitor to various acids and bases. Inhibitor precipitated with trichloroacetic acid (5%, w/v) was fully active when redissolved in Tris buffer.

DISCUSSION

The structure of the inhibitor is unusual. There is little or no methionine but 47 or more residues of half-cystine. This number is a minimum estimate (16) derived from amino acid analysis. Most of it is found as cystine (Fig. 4) and the resulting disulfide bonds probably confer stability to heat and to acid (Table VI) as well as sensitivity to dithiothreitol. We have never been able to recover inhibitor activity after dithiothreitol treatment.

The interaction of the inhibitor with the cyclic nucleotide phosphodiesterase changes the K_m of the enzyme without greatly affecting V_{max} , whether the substrate is cAMP or cGMP (12). The interaction is nearly irreversible, with a K_d of 10^{-10} M or less. The interaction is not salt-sensitive and is slowed but not stopped in the presence of high levels of cAMP.

We have shown that concanavalin A interferes with the inhibitor-enzyme complex formation by binding to the enzyme. This inhibition of complex formation is probably due to steric hindrance. Conditions which prevent binding of concanavalin A to the enzyme, e.g. periodate treatment of the enzyme or the presence of α -methyl-D-mannoside, still allow inhibitor-enzyme interaction. Based on the results obtained following periodate oxidation of the inhibitor and of the enzyme it seems unlikely that the oligosaccharide moieties of inhibitor and enzyme are critical to their interaction.

Our conclusion that the stoichiometry of the enzyme-inhibitor interaction is 1:1 is based on gel filtration data and, therefore, on the assumption that complex formation causes no massive changes in molecular shape.

We are not aware of any similar cyclic nucleotide phosphodiesterase inhibitory protein, with the possible exception of a protein which has been partially purified from the rod outer segment of the bovine eye (21).

The inhibitor of cyclic nucleotide phosphodiesterase is of interest because of the role that it plays during chemotaxis and because the cells regulate its production during growth and development. The experiments shown in Fig. 1 define the times and conditions during which the cells produce large amounts of inhibitor. We do not know whether the induction of inhibitor seen in stationary phase and in buffer is the result of new mRNA synthesis, nor do we know the level of regulation of the suppression of inhibitor secretion brought about by exogenous cAMP (2-5). The purification and physical characterization of the inhibitor make approaches to these questions feasible.

It is now possible to explore the biological role of the inhibitor and the extracellular cyclic nucleotide phosphodiesterase. Since some of the events involved in chemotaxis are

extracellular, the purified inhibitor can be added to fields of aggregating cells and the effect on aggregation observed directly. If extracellular cyclic nucleotide phosphodiesterase is essential to aggregation, a supposition which has been questioned (22), adding a large excess of inhibitor to aggregating cells should block or delay aggregation. The K_m of the complex, which we know forms rapidly, is 2 mM, far above the micromolar levels observed during aggregation (23). Experiments of this type, to be reported elsewhere, have shown that excess inhibitor delays the aggregation of cells at low but not at high density. This is consistent with the results of Wier (24) who showed that cells at a low density preferentially make extracellular phosphodiesterase, while at high density a majority of phosphodiesterase activity is membrane-bound and, therefore, insensitive to inhibitor.

Acknowledgments—We would like to thank Dietmar Kappes for the excellent assistance in some experiments, Mike Brenner for helpful discussions and critical reading of the manuscript, and Harry Orr for assistance with the amino acid analysis. Alice Blatchley and Julie Otto deserve special thanks for typing the manuscript.

REFERENCES

- Loomis, W. F. (1979) *Dev. Biol.* **70**, 1-12
- Klein, C. (1975) *J. Biol. Chem.* **250**, 7134-7138
- Klein, C., and Darmon, M. (1977) *Nature* **268**, 76-78
- Tsang, A. S., and Coukell, M. B. (1977) *Cell Differ.* **6**, 75-84
- Yeh, R. P., Chan, F. K., and Coukell, M. B. (1978) *Dev. Biol.* **66**, 361-374
- Gerisch, G., Malchow, D., Riedel, V., Müller, E., and Every, M. (1972) *Nature New Biol.* **235**, 90-92
- Riedel, V., and Gerisch, G. (1971) *Biochem. Biophys. Res. Commun.* **42**, 119-124
- Riedel, V., Malchow, D., Gerisch, G., and Nägele, B. (1972) *Biochem. Biophys. Res. Commun.* **46**, 279-287
- Eitle, E., and Gerisch, G. (1977) *Cell Differ.* **6**, 339-346
- Dicou, E., and Brachet, P. (1979) *Biochem. Biophys. Res. Commun.* **90**, 1321-1327
- Tsang, A. S., and Coukell, M. B. (1979) *Eur. J. Biochem.* **95**, 419-425
- Kessin, R. H., Orlow, S. J., Shapiro, R. I., and Franke, J. (1979) *Proc. Natl. Acad. Sci. U. S. A.* **76**, 5450-5454
- Toorchen, D., and Henderson, E. J. (1979) *Biochem. Biophys. Res. Commun.* **87**, 1168-1175
- Orlow, S. J., Shapiro, R. I., Franke, J., and Kessin, R. H. (1981) *J. Biol. Chem.* **256**, 7620-7627
- Segrest, J. P., and Jackson, R. L. (1972) *Methods Enzymol.* **28**, 54-63
- Blackburn, S. (1978) in *Amino Acid Determination* (Blackburn, S., ed) pp. 12-14, Dekker, New York
- Spiro, R. G. (1966) *Methods Enzymol.* **8**, 3-5
- Technical Bulletin 1069* (1979) Bio-Rad Laboratories, Richmond, CA
- Greco, W. R., and Hakala, M. T. (1979) *J. Biol. Chem.* **254**, 12104-12109
- Gottschalk, A. (Editor) (1972) *Glycoproteins* Vol. 5, Part A, p. 331, BBA Library, Elsevier, Amsterdam
- Dumler, I. L., and Etingof, R. N. (1976) *Biochim. Biophys. Acta* **429**, 474-484
- Gerisch, G. (1976) *Cell Differ.* **5**, 21-25
- Brenner, M. (1978) *Dev. Biol.* **64**, 210-223
- Wier, P. W. (1977) *Differentiation* **9**, 183-191

Additional references will be found on p. 7637.

Supplementary Material to The Cyclic Nucleotide Phosphodiesterase Inhibitor Protein of *Dicoryctes* *discoideum*: Purification and Characterization, by Jakob Franke and Richard H. Kessin.

MATERIALS AND METHODS

Culture conditions: The axenic strain, AX3, was grown in HL5 medium (1) with 700 ml of liquid medium in 2L erlenmeyer flasks on a gyratory shaker (160 rpm and 21-22°C). The doubling time was 10-11 hours. When the cultures entered stationary phase ($1.5 - 2.0 \times 10^7$ cells/ml) they were harvested by centrifugation (1 min; 1000 x g). The stationary phase supernatant was used as a source of inhibitor in some experiments.

Developmental conditions: To initiate starvation, the cells were washed twice with 17M $\text{Na}_2\text{HPO}_4/\text{KH}_2\text{PO}_4$ (pH 6.0) containing 0.5M CaCl_2 . The washed cells were resuspended at 4×10^6 /ml in the above buffer and 500-600 ml of the cell suspension were incubated at 21-22°C in 2L erlenmeyer flasks on a gyratory shaker at 160 rpm. After 13 hours of shaking in buffer, the cells were removed by centrifugation and the supernatant used as a source of inhibitor. All of the following operations were carried out at 4°C unless otherwise described.

Purification of the inhibitor:

1. Ammonium sulfate precipitation: 435 g of ammonium sulfate were gradually added per liter of supernatant (2.7 M). After addition of $(\text{NH}_4)_2\text{SO}_4$, the solution was left at 4°C for at least 2 hours before the precipitate was collected by centrifugation (1 hr; 13,700 x g). The pellet was resuspended in about 1/20 the original volume in 2.5M ammonium sulfate in Tris (50mM Tris-HCl; pH 7.4 at 35°C). This suspension was centrifuged again (15 min; 27,000 x g) and the pellet dissolved in Tris and then dialyzed for 24 hours against two changes of 2L of the same buffer. Following dialysis, the inhibitor solution was centrifuged (30 min; 48,000 x g) and the supernatant was retained. The pellet was resuspended in 1.5M KCl (to half the volume of the supernatant) and recentrifuged. The supernatants were combined and heated for 20 min at 80°C to destroy phosphodiesterase, cooled in ice, and clarified by centrifugation (30 min; 48,000 x g).

2. Gel filtration: The dialyzed and heated ammonium sulfate precipitate was applied to a Sepharyl S-200 column equilibrated with Tris. Depending on the size of the preparation one of two columns was used. The smaller had a bed volume of 420ml (84 x 2.5 cm) and was eluted at a flow rate of 35 ml/hr. A larger column with a bed volume of 1.55L (88 x 4.7 cm) was eluted at a flow rate of 50 ml/hr. Sample size never exceeded 4% of the bed volume. Inhibitor activity eluted from each column with a K_{av} of 0.24.

$$(1) \quad K_{av} = \frac{V_e - V_o}{V_t - V_o}$$

3. Concanavalin A-Sepharose 4B chromatography: Following gel filtration the fractions containing inhibitor activity were pooled and applied to a Concanavalin A affinity column. The column was prepared and regenerated by washing with two bed volumes of 20% (w/v) α -methylmannoside in Tris. The column was left overnight and then washed with another bed volume of 20% α -methylmannoside, followed by 100 bed volumes of distilled water and 2 bed volumes of Tris containing 1M NaCl and 1mM each of CaCl_2 , MgCl_2 , and MnCl_2 . The column was equilibrated with Tris buffer before use. When the starting preparation was stationary phase supernatant, about 50,000 units could be applied per ml of bed volume, while if the starting material was supernatant from developing cells, 300,000 units per ml of bed volume could be applied without overloading the column. Following binding of the inhibitor, the column was washed with 2 bed volumes each of Tris, Tris containing 1M NaCl, and Tris containing 1M α -methylmannoside. The column was left overnight and then eluted with 4 bed volumes of 1M α -methylmannoside followed by 2 bed volumes of Tris plus 2% α -methylmannoside. The eluates with the highest specific activity were pooled.

4. Ion exchange chromatography: The pooled fractions were applied to a DE-52 column equilibrated with Tris. About 10^6 units could be applied to a 1ml column (0.7 x 2.5cm) and eluted with a linear gradient of 0-0.5M NaCl in Tris buffer (80ml total). The fractions containing activity eluted between 0.05M and 0.2M NaCl. They were pooled and concentrated by ultrafiltration through an Amicon PM-10 membrane under nitrogen.

5. Gel filtration on Bio-Gel A-1.5M: The concentrate (2.5 ml) was applied to a Bio-Gel A-1.5M column (46 x 1.8 cm, bed volume 115ml, $V_o = 69$ ml) and eluted with Tris at a flow rate of 5 ml/hr. Fractions with constant specific activity ($K_{av} = 0.48$) were pooled and concentrated by ultrafiltration.

Assay of Inhibitor:

The inhibitor activity was measured by a modification of the method of Riedel et al (2). One unit of inhibitor is the quantity of inhibitor that reduces 2 units of cyclic nucleotide phosphodiesterase to one unit. The stock reagents for the spectrophotometric assay were: (i) cyclic nucleotide phosphodiesterase which had been partially purified to a specific activity of 12,000 units/mg protein; 350 units/ml in Tris buffer (stable for at least 6 months when stored at -20°C), (ii) coupling enzymes consisting of 16 units of alkaline phosphatase and 10 units of adenosine deaminase per ml of Tris buffer also containing 10mM MgCl_2 (stable for at least 1 week at 4°C) and, (iii) 1mM cAMP in Tris buffer. The assay was initiated by mixing 10 μ l of phosphodiesterase (3.5 units) with the inhibitor sample in a total volume of 0.925 ml of Tris and incubating for 15 minutes at 35°C. The remaining phosphodiesterase activity was then measured spectrophotometrically after adding 25 μ l of coupling enzymes and 50 μ l of 1mM cAMP by following the rate of decrease of $A_{265\text{nm}}$ at 35°C.

Sometimes a radiometric assay was used. 80 μ l of the inhibitor-enzyme mixture were transferred to 10 x 75 mm tubes containing 10 μ l of BSA solution (1 mg/ml). The reaction was initiated by adding 10 μ l of 0.5mM [^3H] cAMP (70 Ci/mol) in Tris buffer. After 10 minutes at 35°C the reaction was terminated by the addition of 10 μ l of trichloroacetic acid (5% w/v) and analysed as described (3). BSA is required to stabilize cyclic nucleotide phosphodiesterase activity during incubation. The results of the two assays correlate closely.

The amount of inhibitor activity detected depends on the source of the cyclic nucleotide phosphodiesterase standard. In initial experiments ammonium sulfate precipitated cyclic nucleotide phosphodiesterase (6500 units/mg protein) was used to standardize the assay and a sigmoidal calibration curve was obtained (Figure 1). Subsequently we found that a linear curve could be obtained if the enzyme fraction was purified further by gel filtration (Figure 1). This enzyme fraction corresponds to the low molecular weight peak described in the accompanying paper (3). The units defined with this enzyme preparation are about 50% higher than those assayed with the ammonium sulfate precipitated enzyme reported previously (4). Cyclic nucleotide phosphodiesterase purified to apparent homogeneity reacts with inhibitor with exactly the same kinetics as the enzyme purified through the gel filtration step.

The assay for cyclic nucleotide phosphodiesterase is described elsewhere (3).

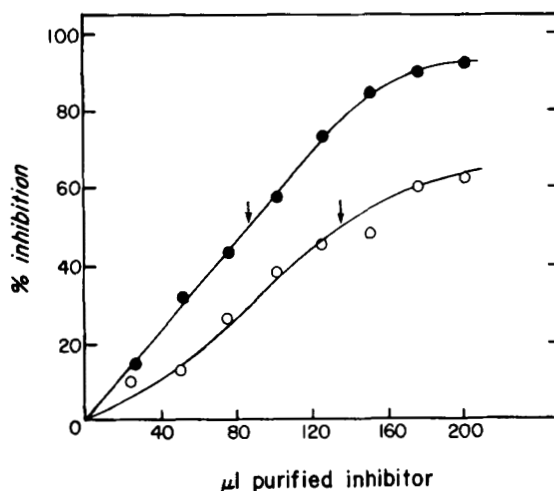


Figure 1: Interaction of the inhibitor with extracellular cyclic nucleotide phosphodiesterase.

○ = cyclic nucleotide phosphodiesterase concentrated by ammonium sulfate precipitation, dialyzed against 2M dithiothreitol in Tris, and subsequently dialyzed against Tris buffer. Contains all forms of extracellular phosphodiesterase (4). The specific activity was 6500 units/mg protein.

◻ = extracellular phosphodiesterase with molecular weight 50,000 obtained by gel filtration on a Sepharyl S-200 column (3). The specific activity was 12,000 units/mg protein.

The inhibitor assays were as described with 3.5 units of phosphodiesterase per tube. The arrows indicate 50% inhibition, or 1.75 units of inhibitor.

Inactivation of the inhibitor by dithiothreitol:

In order to analyze the interaction of the inhibitor and the cyclic nucleotide phosphodiesterase, the optimal conditions for the reactivation of the inhibited enzyme were defined. A sample of purified low molecular weight cyclic nucleotide phosphodiesterase (140,000 units/mg protein) was incubated with a slight excess of purified inhibitor in Tris buffer. Aliquots were mixed with varying concentrations of dithiothreitol, incubated for one hour at 35°C and 16 hours at 4°C, and the remaining enzyme activities determined. Since dithiothreitol has a secondary effect as an inhibitor of phosphodiesterase activity (3) the same experiment was done on samples without inhibitor to be able to correct for this effect. The results are shown in Figure 2 and reveal increasing activation to a plateau at 2 mM dithiothreitol. In a separate experiment, it was determined that an incubation of 60 min at 35°C in the presence of 2mM dithiothreitol was necessary and sufficient for full reactivation of cyclic nucleotide phosphodiesterase (results not shown).

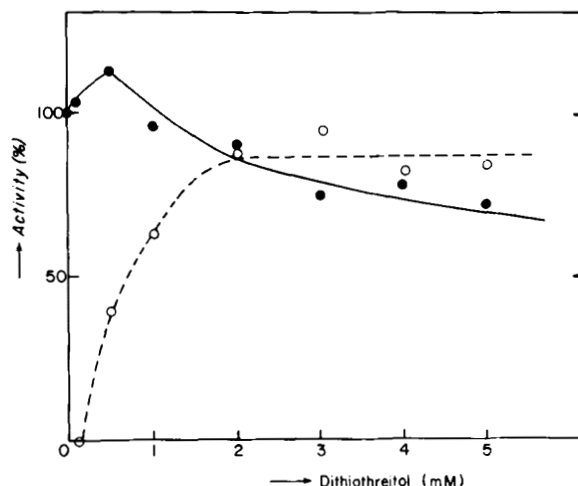


Figure 2: Reactivation of inhibited phosphodiesterase by dithiothreitol.

After treatment with the concentrations of dithiothreitol shown, 50 μ l aliquots were assayed for remaining enzyme activity by the spectrophotometric assay method.

● = Activities recovered when no inhibitor was present.

○ = Activities recovered when inhibitor was present. These values have been corrected for the inhibition of enzyme activity caused by dithiothreitol.

Polyacrylamide gel electrophoresis:

Gel electrophoresis was performed in 5x125mm glass tubes in a Polyanalyst apparatus (Buchler Instruments) or in a model SE-500 slab gel apparatus (Hofer Instruments). Slab gels were 1.5 mm thick. All running gels were 10cm long and spacer gels were about 1.5 cm. For electrophoresis under non-denaturing conditions, the discontinuous pH 8.9 buffer system described by Davis (5) was used. Running gels were prepared with 2.6% cross linking BIS(acrylamide +BIS) and polymerized with 0.07% ammonium persulfate and 0.06% TEMED. Spacer gels contained 2.5% acrylamide and 0.6% BIS and were photopolymerized with riboflavin (5 µg/ml). Samples were made 10% (w/v) sucrose and 0.01% bromophenol blue. Electrophoresis was carried out with a current of 2.5 mA/tube for the first half hour after which the current was raised to 4mA per tube. SDS gel electrophoresis was done using the pH8.9 buffer system described by Laemmli (6). Gels were prepared with 3% cross linking and polymerized with 0.1% ammonium persulfate and 0.05% TEMED. Spacer gels were 3% polyacrylamide with 3% cross linking. Samples in Tris were made 1% SDS and 2% 2-mercaptoethanol and were heated for 2 min in a boiling water bath. Sucrose and bromophenol blue were added, and electrophoresis was carried out with a constant current of 3mA/tube or 30mA per slab gel.

Isoelectrofocusing: Gel mixtures containing 4% (w/v) acrylamide, 0.2% (w/v) BIS, 10% (v/v) glycerol, 1.5% (w/v) ampholytes and 0.05% ammonium persulfate were degassed. TEMED was added (0.05% final concentration) and the gels were polymerized in 5x125mm tubes. After polymerization, the serum caps at the bottom of the tubes were replaced with dialysis membrane. Samples were prepared by adding glycerol (10%) and pH 3-10 ampholytes (2%) and introduced with a microsyringe under a 50mm cushion of 7.5% glycerol and 2% ampholytes. Focusing was done in the Polyanalyst apparatus at 5°C. The lower reservoir contained 600ml of 10mM phosphoric acid and the upper reservoir contained 300 ml of 20mM sodium hydroxide. The focusing was started with a constant current of 2.5mA/tube until the voltage reached 500V; focusing was continued with constant voltage for 2-3 hours by which time the current was less than 0.4mA/tube. pH gradients were determined by elution of 1cm slices in 2ml of 10mM KCl; the pH of the eluates was measured at room temperature. A number of colored pH markers were included in separate gels to provide reference points. Of these, hemoglobin focused between pH 7.4 and 6.9, ferritin between pH 4.3 and 4.6, Aniline Blue between pH 4.2 and 3.9, and Fast Green between pH 3.0 and 2.7.

Gels were stained by incubating them for 15 minutes at 60°C in a solution containing 0.1% Gomaleic Brilliant Blue R-250, 3% (v/v) sulfosalicylic acid, 10% (v/v) trichloroacetic acid, and 30% (v/v) methanol. Gels were destained in an H₂O, ethanol, acetic acid mixture (8:1:1) and stored in 7% acetic acid. Glycoproteins were stained according to the PAS method (7) or with Alcian Blue (8) and the stained gels stored in 7% acetic acid.

Protein was determined by the method of Lowry et al (9) with BSA as standard, or by the optical procedure of Warburg and Christian (10).

Amino acid analysis: Purified inhibitor was precipitated with 90% acetone (v/v) containing 1% trichloroacetic acid (w/v) and the pellet was washed with acetone. The dried pellet was redissolved in constant boiling 5.7 N HCl and phenol was added to a final concentration of 3% (v/v). Aliquots were flushed with N₂, sealed in ampoules under vacuum and hydrolyzed for 24, 48 and 72 hours at 120°C. The hydrolyzed samples were analyzed in a Beckman model 121-MB amino acid analyzer. Cysteine and tryptophan were not determined.

Materials:

[2,8-³H] cAMP (52 Ci/mmol) was purchased from Amersham; [3H] acetic acid (193 Ci/mol) from NEN; Concanavalin A - Sepharose 4B, Sephacryl S-200, Pharmalyse 3-10, and thyroglobulin were obtained from Pharmacia; Ultragel ACA6 and Ampholine 3.5-5, 4-6, and 9-11 were from LKB; Affi-Gel Blue (100-300 mesh), Bio-Gel A-1.5M (100-200 mesh), acrylamide, BIS, SDS, and TEMED were bought from Bio-Rad; alkaline phosphatase (calf intestine; 440 U/mg) was from Boehringer-Mannheim; DE-52 from Whatman; 5.7 N constant boiling HCl was obtained from Pierce; Alcian Blue from Eastman; Schiff Reagent, o-lactone, and Scintiverse were from Fisher; deoxyribonuclease I, phosphorylase A, pyruvate kinase, and purified soybean trypsin inhibitor were from Worthington. The following biochemicals were obtained from Sigma: adenosine deaminase (type I; 200 U/mg), 5'-AMP, cAMP, Aniline Blue (Methyl Blue), Amido Black 10B, BSA (fraction V), ovalbumin (chicken egg; grade V), orcinotryptophan, L-tyrosine, dithiothreitol (reduced and oxidized), dithioerythritol, EDTA, EGTA, Fast Green FCF, ferritin (type I), bovine γ-globulin (Cohn fraction II), hemoglobin (bovine; type I), concanavalin A (type IV), iodoacetic acid, 2-mercaptoethanol, α-methyl-D-mannoside (grade III), Ponceau S, ribonuclease A (type I-A), myo-inositol, and Tris.

RESULTS

Inhibitor purification: The steps used to purify inhibitor from the extracellular supernatant of starving cells are listed in table 1.

Table 1

Purification of inhibitor from the supernatant of cells starved in phosphate buffer.

Purification Step*	Volume (ml)	Units/ml	Units/mg	Yield (%)
Initial supernatant**	3760	540	2570	100
(NH ₄) ₂ SO ₄ ***	39	43,000	6230	83
Sephacryl S-200	102	11,400	125,000	57
ConA-SepharoseB	15	58,600	315,000	43
DE-52	9	72,000	407,000	32
Bio-Gel A-1.5 M	12	42,400	424,000	25

* The details of all purification steps are described in Methods.

** Inhibitor activity was assayed following dialysis of an aliquot of the initial supernatant against Tris, and heating 15 min at 80°C.

*** (NH₄)₂SO₄ refers to material following heating at 80°C in the presence of 0.5M KCl.

Ammonium sulfate precipitation increased the specific activity about 3 fold and usually yielded 80% recovery of inhibitor activity. In order to recover inhibitor as a single peak on a Sephacryl S-200 column, ammonium sulfate precipitates were heated to 80°C in the presence of 0.5 M KCl. If dialysed but unheated material was fractionated on a Sephacryl column, two inhibitor peaks emerged, one with a K_{av} of 0.09 and the other with a K_{av} of 0.30. Heating in the presence of 0.5M KCl converted the high molecular weight peak entirely to the low molecular weight form and improved the purification (table 2).

Table 2

The effects of heating and salts on the gel filtration properties of the inhibitor.

Treatment	Peak I (M_r = 105,000)			Peak II (M_r = 46,000)		
	% of recovered units	Units/mg	A_{260}/A_{280}	% of recovered units	Units/mg	A_{260}/A_{280}
A: 80°C-KCl	73	5800	1.46	27	4907	1.55
B: 80°C-KCl	35	3500	1.20	65	29,300	1.90
C: 80°C-KCl	0	0	--	100	66,400	1.14

Ammonium sulfate precipitate was prepared from the supernatant of cells starved in phosphate buffer. Following dialysis aliquots were: A) centrifuged (15 min - 27,000xg); B) heated at 80°C for 15 min and centrifuged; C) made 0.5M KCl by the addition of one half volume of 1.5 M KCl, heated at 80°C, and centrifuged.

The samples were analyzed on a Sephacryl S-200 gel filtration column (bed volume 118 ml; bed height 48 cm) equilibrated with Tris. The flow rate was 12 ml per hr. The void volume was 47 ml as determined with Blue Dextran, and the column was calibrated with BSA (M_r = 67,000), ovalbumin (M_r = 45,000), and chymotrypsinogen A (M_r = 25,000). Column fractions were heated for 20 min at 80°C before being assayed for inhibitor activity. Without this treatment, the high molecular weight material could not be detected. The low molecular weight peak could be detected with and without heat treatment of the column fractions.

Sephacryl S-200 gel filtration increased the specific activity of the inhibitor about 3 fold. Complete elution required an equilibration of several hours in 1% α-methylmannoside. Ion exchange chromatography increased the specific activity approximately 1.3 fold, leaving minor contaminants which could be removed by a second gel filtration step. A final specific activity of about 440,000 units/mg of protein was obtained whether protein concentrations were determined with the Lowry method or by A_{260}/A_{280} ratios. The same purification scheme has also been used to purify inhibitor from the supernatant of an early stationary phase culture to apparent homogeneity (table 3).

Table 3

Purification of inhibitor from the supernatant of cells in early stationary phase culture.

Purification Step*	Units/mg protein	Yield (%)
Initial sup.	130	100
(NH ₄) ₂ SO ₄	510	100
Sephacryl S-200	2720	70
ConA-Sepharose 4B	52,800	62
DE-52	325,000	34
Bio-Gel A-1.5M	450,000	14

*See table 1.

References

1. Franke, J., and Kessin, R.H. (1977) Proc. Nat. Acad. Sci. USA, **74**, 2157-2161.
2. Riedel, V., Maichow, O., Gerlich, G., and Nägele, B. (1972) Biochem. Biophys. Res. Comm. **46**, 279-287.
3. Orlov, S.J., Shapiro, R.I., Franke, J., and Kessin, R.H. Accompanying paper.
4. Kessin, R.H., Orlov, S.J., Shapiro, R.I., and Franke, J. (1979) Proc. Nat. Acad. Sci. USA, **76**, 5450-5454.
5. Davis, B.J. (1964) Annals N.Y. Acad. Sci. **121**, 404-427.
6. Laemmli, U.K. (1970) Nature, **227**, 680-685.
7. Zacharias, R.M., Zell, T.E., Morrison, J.H., and Woodlock, J.J. (1969) Anal. Biochem. **30**, 148-152.
8. Ward, A.H., and Michos, G.A. (1972) Anal. Biochem. **49**, 607-609.
9. Lowry, O.H., Rosebrough, N.J., Farr, A.L., and Randall, R.J. (1951) J. Biol. Chem. **193**, 265-275.
10. Warburg, O., and Christian, W. (1941) Biochem. Z. **310**, 384-393.